Fiscal Year:	FY 2025	**Task Last Updated:**	FY 10/16/2024
PI Name:	Lybrand, Rebecca Ph.D.		
Project Title:	Growing Food on Mars: Determining the Impact of Radiation, Atmospheric Composition, and Rock Substrate on Plant Growth in a Space Rock Garden Experiment		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	95616-5270	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBPS E.9: Plant Studies
Start Date:	12/15/2022	**End Date:**	12/13/2024
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/13/2024 per NSSC information (Ed., 10/24/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rodrigues, Jorge Mazza Ph.D. (University of California, Davis) Melotto, Maeli Ph.D. (University of California, Davis) Zaharescu, Dragos Ph.D. (University of California, Davis)		
Grant/Contract No.:	80NSSC23K0371		
Performance Goal No.:			
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Task Description:	1. SCIENCE GOALS AND OBJECTIVES. As human civilization expands outside of its terrestrial cradle to explore the Moon and Mars, the sources and delivery of nutrients for long-duration missions must be identified and refined. The importance of using local mineral resources for sustaining life, and the bioengineering of such environments remain at the vanguard of sustainable human space exploration. Our overarching goal is to test how ionizing radiation, atmospheric composition, and rock substrate constrain and influence plant growth in deep space exploration, specifically the maintenance of plants in Lunar and Martian environments. This Early Career Investigation (ECI) will produce new publishable findings that integrate how food plants interactively respond to spaceflight stressors (carbon dioxide / CO2 and radiation) and environmental constraints imparted by basalt rocks containing different morphological and elemental arrangements that serve as nutrient sources for plants and microbes in Mars-relevant environments. Once constructed, the Space Rock Garden Experiment (SRGE) will serve as the framework for performing additional plant studies experiments to be proposed through full-ground based proposals and future International Space Station (ISS) flight experiments. We will achieve three objectives: 1) Develop and construct the SRGE, an integrated experimental system capable of controlling the mineral substrate, water, atmospheric and ultraviolet (UV) radiative conditions, and the presence of plants and microbes; 2) Identify how the flux of short wavelength (UV-B) radiation and atmospheric composition influence the rock weathering environment (e.g., nutrient elements compartmentalization), therefore assessing how coupled atmospheric and stellar energy sources influence the formation and habitability of incipient soils; 3) Integrate tomato and N-fixing plant genotypes, arbuscular mycorrhiza, and associated microbiota into the SRGE to assess how rock properties affect the growth and development of plants as viable crops for deep space exploration under increased CO2 and UV-B radiation. 2. METHODOLOGY. We will design, construct, and test the SRGE to simulate plant growth and microbe-mineral interactions under atmospheric and radiation scenarios relevant to Martian landscape. We will assess plant and microbial stress indicators in combination with biogeochemical analyses of major and trace elements in mineral, water and biomass pools. Micro-XCT (X-Ray Computed Tomography) will be used to assess plant root architecture, pore space morphology, and the biogeochemical indicators required to support complex plant life. We performed a pilot study using basalt rock substrates sampled from Mars analog sites in Iceland and confirmed that: i) tomato and lentil plants successfully co-germinated and grew together in basalt rock substrates under ambient conditions; ii) DNA was can be extracted from fresh basalt rock substrates, indicating that the rock materials are capable of hosting microbial life; and iii) a microXCT approach successfully differentiated dense mineral particles, water-filled pores, air-filled pores, and roots from tomato plants grown in the basalt rock substrates.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Rock weathering replenishes nutrients in the environment, shapes geochemical carbon sequestration, and drives physical, chemical, and/or biological processes at a multitude of scales. Globally, lithology type and distribution, climate, and ecosystem activity are predictors of mineral weathering rates and long-term carbon cycling. Our research is examining the intricacies of how contrasting crystalline and glassy basaltic rock substrates weather and release mineral-bound elements that are essential to life using controlled environmental chambers through the Space Rock Garden Experiment (SRGE). We are also examining how morphological and elemental differences in basalt rock composition impacts food plant production under both Earth and Mars atmospheric conditions, which is expanding understanding of how rock substrate composition influences plant growth. Our work is advancing knowledge of sustainable food production using natural (rock-based) fertilization through basalt weathering. Carbon-capture through mineral weathering is another important secondary effect.
Task Progress:	A main objective for our one-year, Early Career Investigation (ECI) grant is to develop and construct the Space Rock Garden Experiment (SRGE), which is an integrated experimental system capable of controlling the mineral substrate, water, atmospheric composition, and UV radiative conditions, and the presence of plants and microbes. Excellent progress has been made on the design, development, and construction of the SRGE. The construction and testing of the SRGE was completed during this reporting period. The construction and testing of the SRGE chambers are top priorities for the project so that we can simulate energetic surface conditions on Mars, both through the current research project and in the submission of upcoming proposals to NASA. Construction and testing of SRGE Environmental Chambers: The construction and testing phase of the project involved a multi-month, iterative process led by Science Principal Investigator (PI) Zaharescu. Science PI Zaharescu organized meetings with experts who offered insight on the construction and testing phase for the SRGE environmental chambers. Science PI Zaharescu also led organizational meetings and discussions with our team to solicit input and to receive any feedback on the process (including the testing of materials, and selection of sensors, among other logistical and technical questions/activities required to advance the project). Science PI Zaharescu and PI Lybrand also worked on purchasing tools and equipment needed for the chamber construction and testing phase, including the sensors, meters, and analysis tools, as well as the purchase of other supplies required to complete the construction of the chambers. Both Science PI Zaharescu and PI Lybrand worked on identifying and preparing a space for the SRGE Environmental Chambers in the laboratory. Science PI Zaharescu also worked extensively to set up the laboratory work space to construct and then to subsequently test the SRGE environmental chambers. Mars analog rock substrate characterization, analysis, and selection for SRGE: Our team employed an electron microprobe analysis technique to continue comparing and contrasting Mars analog rock substrate materials collected from Mars analog sites to characterize the substrates included as the primary substrates in the Space Rock Garden Experiments. Preliminary data sets are suggesting geochemical similarities between Mars rock measurements and Mars analog rock materials used in our experiments. The electron microprobe analysis approach also produced micrographs, which provided microscale imagery of the Mars analog rock substrates, including glassy and crystalline substrates, that are being examined in the experiments. Construction of SRGE Environmental Chambers: The construction and testing of the Mars simulation chambers was performed in iterative phases using an innovative design thinking method, and a versatility-esthetics centered approach, that included: 1) Ideation, 2) Mapping, 3) Design Block, 4) Implementation, and 5) Testing. Identifying and testing Mars-relevant radiation and lighting conditions: Science PI Zaharescu implemented the findings from the compilation of an extensive database to select the parameters for the Martian radiation component of the SRGE. The resulting database integrated data from 15+ published papers and data sets from Mars rovers and satellites

	on the Martian radiation environment. Science PI Zaharescu also organized several meetings to discuss the findings of the literature review and the subsequent selection of the radiation lamps and conditions with university experts and NASA scientists. Science PI Zaharescu also conducted an exhaustive literature review and compiled 15+ papers on plant use wavelengths that would be relevant to Mars for use in selecting lighting system components. Science PI Zaharescu tested the radiation scenarios by measuring and producing UV-C radiance maps on the chamber platform that was used for Mars radiation treatments on the Mars analog basalt substrates. Identifying and testing the Mars atmospheric component of the SRGE: Science PI Zaharescu conducted a thorough review of literature on Martian atmospheric compositions and compiled 20+ papers as a result. We selected an atmospheric composition that has a high-degree of similarity with the Martian atmosphere at Gale Crater. We worked with the university and purchasing department to order custom Mars atmosphere gas mixture tanks composed of CO2, N2, Ar, O2, and CO. Once the construction of the surge chambers was complete and the Mars custom gas mixtures were acquired, Science PI Zaharescu performed intensive tests of the chambers to confirm that the chambers maintained the high CO2, low O2 atmospheric conditions that are relevant to Mars. The tests were a success and the continuous, in-situ oxygen sensors demonstrated that the Mars atmospheric conditions were maintained for the duration of the multiweek weathering experiments that were subsequently performed in the chambers once the testing of the Mars radiation and atmospheric components was complete. Abiotic Weathering and Plant Growth Experiments: The completion of the construction and testing of the SRGE environmental chambers was finished during the current reporting period. Science PI Zaharescu set up and oversaw multiple abiotic weathering experiments including the multiweek exposure of crystalline and glassy basalt substrates to Mars atmospheric conditions and the subsequent exposure of basalt substrates to Mars UV-C radiation conditions. Finally, the plant growth experiments are presently underway with an estimated completion date of Fall 2024.
Bibliography Type:	Description: (Last Updated: 07/30/2025)
Abstracts for Journals and Proceedings	Lybrand RA, et al. "Connecting soils to life on Earth and beyond." SSSA 2024 (Soil Science Society of America 2024) Bouyoucos Summer Conference, San Juan, Puerto Rico, June 10-12, 2024. Abstracts. SSSA 2024 (Soil Science Society of America 2024) Bouyoucos Summer Conference, San Juan, Puerto Rico, June 10-12, 2024. Plenary Talk. https://www.sacmeetings.org/speakers/ , Jun-2024
Abstracts for Journals and Proceedings	Rodrigues J, et al. "The role of natural organic matter and humic substances in achieving sustainable development goals." 22nd Meeting of the IHSS (International Humic Substances Society), Rimini Italy, August 25-30, 2024. Abstracts. 22nd Meeting of the IHSS (International Humic Substances Society), Rimini Italy, August 25-30, 2024. , Aug-2024
Abstracts for Journals and Proceedings	Rodrigues J, et al. "The soil, plant, and human continuum." Universita' degli Studi di Firenze, Florence, Italy, June 05, 2024. Presentation. Universita' degli Studi di Firenze, Florence, Italy, June 05, 2024. , Jun-2024
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Abstracts for Journals and Proceedings	Lybrand RA, et al. "Microscale analyses of mineral particles and applications to dust research." University of California Dust Symposium: Beyond the Haze, University of California Davis, Davis, CA, May 15, 2024. Presentation. University of California Dust Symposium: Beyond the Haze, University of California Davis, Davis, CA, May 15, 2024. , May-2024
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